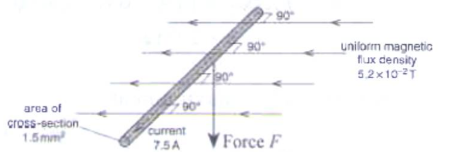
2(c)(iii)	The efficiency of power transfer increases as R increases from $4\,\Omega$ to $10\,\Omega$, since P_R decreases at a lower rate than P_T with increasing R.
3(a)(i)	
3(a)(ii)	$F = BIL\sin 90°$ $\dfrac{F}{L} = BI$ $= 5.2\times10^{-2}(7.5)$ $= 0.39\ \mathrm{N\,m^{-1}}$
3(b)(i)	Volume per unit length of wire = $1.5\times10^{-6}\ \mathrm{m^2}$ Number of electrons present per unit length of wire = $(7.8\times10^{28})1.5\times10^{-6}$ $=1.17\times10^{23}$ Since force per unit length of wire is 0.39 N, Force acting on each electron = $\dfrac{0.39}{1.17\times10^{23}}$ $=3.3\times10^{-24}$ N
3(b)(ii)	Using $F = Bqv\sin 90°$ $3.333\times10^{-24} = 5.2\times10^{-2}\times1.6\times10^{-19}v$ $v = 4.0\times10^{-4}\ \mathrm{m\,s^{-1}}$
3(c)	The electric field in the wire, produced by the battery, would cause all the electrons to move almost simultaneously. Although the drift speed is low, the high number density of free electrons means a large number of electrons per unit time colliding with the ionic lattice in the filament, causing thermal vibrations. This results first in thermal energy, then light being emitted.
4 (a)	Faraday's law of electromagnetic induction states that the induced e.m.f. in a